

Unit 02: Average

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Objective

The key concepts explained in this unit, including area a learner is expected to learn and master after going through the unit are: -

- Understand the average of all numbers.
- Understand what Weighted Average are

Introduction

Find the average of all prime numbers between 30 and 50? Sol: There are five prime numbers between 30 and 50.

They are 31,37,41,43 and 47.

Therefore the required average = $(31+37+41+43+47)/5 \Leftrightarrow 199/5 \Leftrightarrow 39.8$.

Whenever we are asked the marks we scored in any examination, we usually tell the marks in terms of percentage that is, taking the percentage of total marks of all subjects. This percentage is called average percentage. Also, in a class, if there are 100 students, instead of knowing the age of individual student, we usually talk about their average age. Introduction Whenever we are asked the marks we scored in any examination, we usually tell the marks in terms of percentage, that is, taking the percentage of total marks of all subjects. This percentage is called average percentage. Also, in a class, if there are 100 students, instead of knowing the age of individual student, we usually talk about their average age. $\frac{3+9+11+15+18+19+23}{7} = \frac{98}{7} = 14$

2.1 Average

$$01 \text{ Average} = \frac{\text{Sum of qualities}}{\text{Number of qualities}}$$

$$02 \text{ Sum of quantities} = \text{Average} \times \text{Number of quantities}$$

$$03 \text{ Number of quantities} = \frac{\text{sum of qualities}}{\text{Average}}$$

Find the average of first 40 natural numbers?

Sol: sum of first n natural numbers = $n(n+1)/2$; So, sum of 40 natural numbers = $(40*41)/2 \Leftrightarrow 820$.

Therefore the required average = $(820/40) \Leftrightarrow 20.5$.

Analytical Skills-I



Example 1: A man purchased 5 toys at rs.200 each, 6 toys at rs250 each and 9 toys at `300 each. Calculate the average cost of 1 toy.

Solution: Price of 5 toys = $200 \times 5 = \text{rs.}1000$

Price of 6 toys = $250 \times 6 = \text{rs.}1500$

Price of 9 toys = $300 \times 9 = \text{rs.}2700$

Total number of toys = $5 + 6 + 9 = 20$

Average price of 1 toy = $\frac{1000+1500+2700}{20}$

$$= \frac{5200}{20} = \text{rs.} 260.$$

Find the average of first 20 multiples of 7?

Sol: Required average = $7(1+2+3+\dots+20)/20 \Leftrightarrow (7*20*21)/(20*2) \Leftrightarrow (147/2)=73.5.$



Example 2: The average marks obtained by 200 students in a certain examination is 45. Find the total marks.

Solution: Total marks

= Average marks $\times$ Number of students

= $200 \times 45 = 900.$

The average of four consecutive even numbers is 27. find the largest of these numbers?

Sol: let the numbers be $x, x+2, x+4$ and $x+6$. then,

$$(x+(x+2)+(x+4)+(x+6))/4=27 \Leftrightarrow (4x+12)/4=27 \Leftrightarrow x+3=27 \Leftrightarrow x=24.$$

Therefore the largest number = $(x+6) = 24+6=30.$



Example 3: Total temperature for the month of September is 840°C . If the average temperature of that month is 28°C , find out the number of days is the month of September.

Solution: Number of days in the month of September

$$= \frac{\text{Total temperature}}{\text{Average temperature}} = \frac{840}{28} = 30 \text{ days.}$$

Short cut methods

01 Average of two or more groups taken together. (a) If the number of quantities in two groups are n_1 and n_2 and their average is x and y , respectively, the combined average (average of all of them put together) is

$$\frac{n_1x + n_2y}{n_1 + n_2}$$

Explanation

Number of quantities in the first group = n_1

Their average = x

$\therefore$ Sum = $n_1 \times x$ Number of quantities in the second group = n_2 Their average = y

$\therefore$ Sum = $n_2 \times y$

Number of quantities in the combined group = $n_1 + n_2$. Total sum (sum of quantities of the first group and the second group) = $n_1x + n_2y$.

∴ Average of the two groups

$$\frac{n_1x + n_2y}{n_1 + n_2}$$

(b) If the average of n_1 quantities is x , and the average of n_2 quantities out of them is y , the average of the remaining group (rest of the quantities) is

$$\frac{n_1x + n_2y}{n_1 + n_2}$$

Explanation

Number of quantities = n_1

Their average = x

∴ Sum = $n_1 \times$ Number of quantities taken out = n_2

Their average = y

∴ Sum = n_2y

Sum of remaining quantities = $n_1x - n_2y$ Number of remaining quantities = $n_1 - n_2$

∴ Average of the remaining group = $\frac{n_1x + n_2y}{n_1 - n_2}$.

There are two sections A and B of a class consisting of 36 and 44 students respectively. If the average weight of section A is 40kg and that of section B is 35kg, find the average weight of the whole class?

Sol: total weight of (36+44) students = (36*40+44*35)kg = 2980kg.

Therefore weight of the total class = (2980/80)kg = 37.25kg.



Example 4: The average weight of 24 students of section A of a class is 58 Kg, whereas the average weight of 26 students of section B of the same class is 60.5 Kg. Find out average weight of all the 50 students of the class.

Solution: Here, $n_1 = 24$, $n_2 = 26$, $x = 58$, and $y = 60.5$.

∴ Average weight of all the 50 students

$$\frac{n_1x + n_2y}{n_1 + n_2}$$

$$= \frac{24 \times 58 + 26 \times 60.5}{24 + 26}$$

$$= \frac{1392 + 1573}{50} = \frac{2965}{50} = 59.3 \text{ kg}$$

Nine persons went to a hotel for taking their meals 8 of them spent Rs.12 each on their meals and the ninth spent Rs.8 more than the average expenditure of all the nine. What was the total money spent by them?

Sol: Let the average expenditure of all nine be Rs. x

Then $12 \times 8 + (x+8) = 9x$ or $8x = 104$ or $x = 13$.

Total money spent = $9x = \text{Rs.}(9 \times 13) = \text{Rs.}117$



Example 5: Average salary of all the 50 employees including 5 officers of a company is rs.850. If the average salary of the officers is rs.2500, find the average salary of the remaining staff of the company. Solution: Here, $n_1 = 50$, $n_2 = 5$, $x = 850$ and $y = 2500$.

∴ Average salary of the remaining staff

$$\frac{n_1x + n_2y}{n_1 - n_2} = \frac{50 \times 850 - 5 \times 2500}{50 - 5}$$

Analytical Skills-I

$$\frac{42500-12500}{45} = \frac{30000}{45}$$

=rs 667(approx)

02 If $\bar{x}$ is the average of $x_1, x_2, \dots, x_n$, then

(b) The averages of $x_1 - a, x_2 - a, x_n - a$ is $\bar{x} - a$.

(c) The average of $ax_1, ax_2, \dots, ax_n$ is $a\bar{x}$, provided $a \neq 0$.

(d) The average of $\frac{x_1}{a}, \frac{x_2}{a}, \dots, \frac{x_n}{a}$ is $\frac{\bar{x}}{a}$, provided $a \neq 0$.



Example 6: The average value of six numbers 7, 12, 17, 24, 26 and 28 is 19. If 8 is added to each number, what will be the new average?

Solution: The new average = $\bar{x} + a$.

The average of 25 result is 18. The average of 1st 12 of them is 14 & that of last 12 is 17. Find the 13th result.

Sol: Clearly 13th result = (sum of 25 results) - (sum of 24 results) = $(18 \times 25) - (14 \times 12) + (17 \times 12) = 450 - (168 + 204) = 450 - 372 = 78$



Example 7: The average of x numbers is $5x$. If $x - 2$ is subtracted from each given number, what will be the new average?

Solution: The new average = $\bar{x} - a$

$$= 5x - (x - 2) = 4x + 2.$$



Example 8: The average of 8 numbers is 21. If each of the numbers is multiplied by 8, find the average of a new set of numbers. Solution: The average of a new set of numbers = $a\bar{x} = 8 \times 21 = 168$

03 The average of n quantities is equal to x . If one of the given quantities whose value is p , is replaced by a new quantity having value q , the average becomes y , then $q = p + n(y - x)$.

The average weight of 10 oarsmen in a boat is increased by 1.8 kg when one of the crew, who weighs 53 kg is replaced by a new man. Find the weight of the new man.

Sol. Total weight increased = (1.8×10) kg = 18 kg.

$\therefore$ Weight of the new man = $(53 + 18)$ kg = 71 kg



Example 9: The average weight of 25 persons is increased by 2 Kg when one of them whose weight is 60 Kg, is replaced by a new person. What is the weight of the new person?

Solution: The weight of the new person

$$= p + n(y - x)$$

$$= 60 + 25(2) = 110 \text{ Kg.}$$

04 (a) The average of n quantities is equal to x . When a quantity is removed, the average becomes y . The value of the removed quantity is $n(x - y) + y$.

(b) The average of n quantities is equal to y . When a quantity is added, the average becomes x . The value of the new quantity is $n(y - x) + y$.



Example 10: The average age of 24 students and the class teacher is 16 years. If the class teacher's age is excluded, the average age reduces by 1 year. What is the age of the class teacher?

Solution: The age of class teacher

$$= n(x - y) + y$$

$$= 25(16 - 15) + 15 = 40 \text{ years.}$$



Example 11: The average age of 30 children in a class is 9 years. If the teacher's age be included, the average age becomes 10 years. Find the teacher's age

. Solution: The teacher's age

$$= n(y - x) + y$$

$$= 30(10 - 9) + 10 = 40 \text{ years.}$$

A batsman makes a score of 87 runs in the 17th inning and thus increases his avg by

3. Find his average after 17th inning.

Sol. Let the average after 17th inning = x . Then, average after 16th inning = $(x - 3)$.

$$\therefore 16(x - 3) + 87 = 17x \text{ or } x = (87 - 48) = 39$$

05 (a) The average of first n natural numbers is $\frac{n+1}{2}$

(b) The average of square of natural numbers till n is $\frac{(n+1)(2n+1)}{6}$

(c) The average of cubes of natural numbers till n is $\frac{n(n+1)^2}{4}$

(d) The average of odd numbers from 1 to n is $\frac{\text{last odd number} + 1}{2}$

(e) The average of even numbers from 1 to n is $\frac{\text{last even number} + 2}{2}$



Example 12: Find the average of first 81 natural numbers.

Solution: The required average

$$= \frac{n+1}{2} = \frac{81+1}{2} = 41$$



Example 13: What is the average of squares of the natural numbers from 1 to 41?

Solution: The required average

$$= \frac{(n+1)(2n+1)}{6} = \frac{(41+1)(2 \times 41 + 1)}{6}$$

$$= \frac{42 \times 83}{6} = \frac{3486}{6} = 581$$



Example 14: Find the average of cubes of natural numbers from 1 to 27.

Solution: The required average

$$= \frac{n(n+1)^2}{4} = \frac{27 \times (27+1)^2}{4}$$

$$= \frac{27 \times 28 \times 28}{4} = \frac{21168}{4} = 5292$$



Example 15: What is the average of odd numbers from 1 to 40?

Solution: The required average

$$= \frac{\text{last odd number} + 1}{2} = \frac{39+1}{2} = 20.$$

Distance between two stations A and B is 778 km. A train covers the journey from A to B at 84 km per hour and returns back to A with a uniform speed of 56 km per hour. Find the average speed of

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the train during the whole journey. Sol. Required average speed = $\frac{(2xy)}{(x+y)}$ km / hr = $\frac{(2 \times 84 \times 56)}{(84+56)}$ km/hr = $\frac{(2 \times 84 \times 56)}{140}$ km/hr = 67.2 km/hr.



Example 16: What is the average of even numbers from 1 to 81? Solution: The required average

$$\frac{\text{last even number} + 1}{2} = \frac{80 + 2}{2} = 41$$

06 (a) If n is odd: The average of n consecutive numbers, consecutive even numbers or consecutive odd numbers is always the middle number.

(b) If n is even: The average of n consecutive numbers, consecutive even numbers or consecutive odd numbers is always the average of the middle two numbers.

(c) The averages of first n consecutive even numbers is $(n + 1)$.

(d) The average of first n consecutive odd numbers is n.

(e) The averages of squares of first n consecutive even numbers is $\frac{2(n+1)(2n+1)}{3}$

(f) The average of squares of consecutive even numbers till n is $\frac{(n+1)(n+2)}{3}$

(g) The average of squares of consecutive odd numbers till n is $\frac{n(n+2)}{3}$

(h) If the average of n consecutive numbers is m, then the difference between the smallest and the largest number is $2(n - 1)$.



Example 17: Find the average of 7 consecutive numbers 3, 4, 5, 6, 7, 8, 9.

Solution: The required average = middle number = 6.



Example 18: Find the average of consecutive odd numbers 21, 23, 25, 27, 29, 31, 33, 35.

Solution: The required average = average of middle two numbers

= average of 27 and 29

$$= \frac{27 + 29}{2} = 28$$



Example 19: Find the average of first 31 consecutive even numbers.

Solution: The required average = $(n + 1) = 31 + 1 = 32$.



Example 20: Find the average of first 50 consecutive odd numbers.

Solution: The required average = n = 50.



Example 21: Find the average of squares of first 19 consecutive even numbers.

Solution: The required average = $\frac{2(n+1)(2n+1)}{3}$

$$= \frac{2 \times 20 \times 39}{3} = \frac{1560}{3} = 520$$



Example 22: Find the average of squares of consecutive even numbers from 1 to 25.

Solution: The required average

$$\begin{aligned} &= \frac{(n+1)(n+2)}{3} = \frac{(25+1)(25+2)}{3} \\ &= \frac{26 \times 27}{3} = \frac{702}{3} = 234 \end{aligned}$$



Example 23: Find the average of squares of consecutive odd numbers from 1 to 31.

Solution: The required average

$$\begin{aligned} \frac{n(n+2)}{3} &= \frac{31 \times (31+2)}{3} \\ &= \frac{31 \times 33}{3} = 341 \end{aligned}$$



Example 24: If the average of 6 consecutive numbers is 48, what is the difference between the smallest and the largest number?

Solution: The required difference

$$= 2(n-1) = 2(6-1) = 10.$$

07 Geometric Mean or Geometric Average.

Geometric mean of $x_1, x_2, \dots, x_n$ is denoted by

$$G.M = \sqrt[n]{x_1 \times x_2 \times \dots \times x_n}$$

Geometric mean is useful in calculating averages of ratios such as average population growth rate, average percentage increase and, so on.



Example 25: The production of a company for three successive years has increased by 10%, 20% and 40%, respectively. What is the average annual increase of production?

Solution: Geometric mean of x, y and $z = (x \times y \times z)^{1/3}$

$$\therefore \text{Average increase} = (10 \times 20 \times 40)^{1/3\%} = 20\%$$



Example 26: The population of a city in two successive years increases at the rates of 16% and 4%, respectively. Find out the average increase in two years.

Solution: In case of population increase, the geometric mean is required.

$\therefore$ Geometric mean of 16% and 4% is

$$= (16 \times 4)^{1/2} \%, \text{ i.e., } 8\%$$

08 Harmonic Mean or Harmonic Average.

Harmonic mean of $x_1, x_2, \dots, x_n$ is denoted by

$$H.M = \frac{1}{\frac{1}{n} \left(\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \right)}$$

Harmonic mean is useful in finding out average speed of a vehicle, average production per day and, so on.



Example 27: A man runs 1 Km at a speed of 15 Km/h and another 1 Km he walks at a speed of 5 Km/h. Find out his average speed in covering 2 Km.

Solution: Harmonic mean is used when distance remains constant and speed varies. Harmonic mean of x and y is $\frac{2}{\frac{1}{x} + \frac{1}{y}}$ or, $\frac{2xy}{x+y}$.

$\therefore$ Average speed for the whole distance

$$= \frac{2 \times 15 \times 5}{15+5} = 7.5 \text{ km/h}$$

09 If a certain distance is covered at a speed of x Km/h and the same distance is covered at a speed of y Km/h, the average speed during the entire journey is

$$\left(\frac{2xy}{x+y} \right) \text{ km/h}$$

Analytical Skills-I



Example 28: If half of the journey is travelled at a speed of 15 Km/h and the remaining half at a speed of 12 Km/h, find out average speed during the entire journey

Solution: The average speed

$$\left(\frac{2xy}{x+y} \right) = \frac{2 \times 15 \times 12}{15 + 12} = \frac{360}{27} = 13\frac{1}{3} \text{ km/h}$$



Example 29: A man goes to a certain place at a speed of 30 Km/h and returns to the original place at a speed of 20 Km/h, find out his average speed during this up and down journey.

Solution: The average speed

$$= \left(\frac{2xy}{x+y} \right) = \left(\frac{2 \times 30 \times 20}{30 + 20} \right) = \frac{1200}{50} = 24 \text{ km/h}$$

10 If a person or a motor car covers three equal distances at the speed of x Km/h, y Km/h and z Km/h, respectively, then for the entire journey average speed of the person or motor car is

$$\left(\frac{3xyz}{xy + yz + zx} \right)$$



Example 30: A train covers the first 160 Km at a speed of 120 Km/h, another 160 Km at 140 Km/h and the last 160 Kms at 80 Km/h. Find out average speed of the train for the entire journey.

Solution: Average speed = $\frac{3xyz}{xy + yz + zx}$

$$\begin{aligned} &= \frac{3 \times 120 \times 140 \times 80}{120 \times 140 + 140 \times 80 + 80 \times 120} \\ &= \frac{360140 \times 80}{16800 + 11200 + 9600} = \frac{4032000}{37600} \\ &= 107\frac{11}{47} \text{ km/h} \end{aligned}$$

11 If a person covers A Km at a speed of x Km/h, B Km at a speed of y Km/h and C Km at a speed of z Km/h, the average speed during the entire journey is

$$\left(\frac{A + B + C}{\frac{A}{x} + \frac{B}{y} + \frac{C}{z}} \right) \text{ km/h}$$



Example 31: A person covers 9 Km at a speed of 3 Km/h, 25 Km at a speed of 5 Km/h and 30 Km at a speed of 10 Km/h. Find out average speed for the entire journey.

Solution: The average speed = $\left(\frac{A+B+C}{\frac{A}{x} + \frac{B}{y} + \frac{C}{z}} \right) \text{ km/h}$

$$\left(\frac{9 + 25 + 30}{\frac{9}{3} + \frac{25}{5} + \frac{30}{10}} \right)$$

$$= \frac{64}{11} = 5\frac{9}{11} \text{ km/h}$$

12 If a person covers A th part of the distance at x Km/h, Bth part of the distance at y Km/h and the remaining C th part at z Km/h, then the average speed during the entire journey is

$$\left(\frac{1}{\frac{A}{x} + \frac{B}{y} + \frac{C}{z}} \right) km/h$$



Example 32: A person covers the first $\frac{1}{4}$ of the distance at 8 Km/h, the next $\frac{3}{5}$ at 6 Km/h and the remaining distance at 15 Km/h. Find the average speed during the entire journey. Solution: The average speed

$$= \frac{1}{\left(\frac{A}{x} + \frac{B}{y} + \frac{C}{z}\right)} = \left(\frac{1}{\frac{1/4}{8} + \frac{3/5}{6} + \frac{3/20}{15}} \right)$$

$$\left[\text{Here, } A = \frac{1}{4}, B = \frac{3}{5} \text{ and } C = 1 - \left(\frac{1}{4} + \frac{3}{5} \right) = \frac{3}{20} \right]$$

$$= \frac{1}{\frac{1}{31} + \frac{1}{10} + \frac{1}{100}} = \frac{3200}{452} = 7 \frac{9}{113} km/h$$



Example 33: A train covers 50% of the journey at 30 Km/h, 25% of the journey at 25 Km/h and the remaining at 20 Km/h. Find the average speed of the train during the entire journey.

Solution: Let the total journey be = 100m. The average speed: = 100 km

$$\left(\frac{100}{\frac{A}{x} + \frac{B}{y} + \frac{C}{z}} \right) = \left(\frac{100}{\frac{50}{30} + \frac{25}{25} + \frac{25}{20}} \right)$$

$$[\text{Here, } A = 50, B = 25 \text{ and } C = 25]$$

$$= \frac{100}{47/12} = \frac{1200}{47} = 25 \frac{25}{47} km/h$$

2.2 Weighted Average

To find the weighted average (or weighted mean) of a data set, multiply each value by its weight and add the products together, then divide by the sum of the weights.

For a set of data x

$1, x_2, \dots, x_n$ with nonnegative weights $w_1, w_2, \dots, w_n$

the weighted average is

$$[x_1w_1 + x_2w_2 + \dots + x_nw_n] / w_1 + w_2 + \dots + w_n.$$

Finding a Weighted Average

A grade point average is a weighted average that gives greater weight to courses that earn more credits. Hailey's grade points are 4.0 in Chemistry, which is worth 4 credits, 3.5 in English, which is worth 3 credits, and 3.7 in Physics, which is worth 2 credits. What is Hailey's grade point average?

SOLUTION

Multiply each grade by its weight. Add the products and then divide by the sum of the weights.

$$4.0(4) + 3.5(3) + 3.7(2) / 4 + 3 + 2 = 33.9 / 9 = 3.7\bar{6}.$$

Analytical Skills-I

Hailey's grade point average is approximately 3.77.

Sometimes the weights associated with data values are numbers between 0 and the sum of the weights is 1. In this case, you can use the following equivalent formula for the weighted average.

Summary

The key concepts learned from this unit are: -

- We have learnt the average of all numbers.
- We have learnt what are Weighted Average.

Keywords

- average
- Weighted Average.

Self Assessment

1. The average of 10 numbers is 7. What will be the new average if each of the numbers is multiplied by 8?
A. 45
B. 52
C. 56
D. 55
2. There are 35 students in a hostel. If the number of students increased by 7, the expenses of the mess were increased by Rs42 per day while the average expenditure per head decreased by Rs 1. Find out the actual expenditure of the mess.
A. Rs 480
B. Rs 440
C. Rs 520
D. Rs 420
3. An aero plane travels 2500 Km, 1200 Km and 500 Km at 500 Km/h, 400 Km, and 250 Km/h, respectively. The average speed is:
A. 420 Km/h
B. 410 Km/h
C. 405 Km/h
D. 575 Km/h
4. The average age of 5 members is 21 years. If the age of the youngest member be 5 years, find out the average age of the family at the birth of the youngest member.
A. 24 years
B. 25 years
C. 20 years
D. 28 years

5. The average weight of 10 students is increased by half a Kg when one of the students weighing 50 Kg is replaced by a new student. Find out the weight of the new student.
- A. 55 Kg
B. 60 Kg
C. 45 Kg
D. 40 Kg
6. The mean marks of 10 boys in a class is 70%, whereas the mean marks of 15 girls is 60%. The mean marks of all the 25 students is:
- A. 64%
B. 60%
C. 55%
D. 52%
7. The average of five consecutive even numbers starting with 4, is:
- A. 6
B. 7
C. 8
D. 7.5
8. The average of 17 numbers is 10.9. If the average of first nine numbers is 10.5 and that of the last 9 numbers is 11.4, the middle number is:
- A. 11.8
B. 11.4
C. 10.9
D. 11.7
9. The sum of three numbers is 98. If the ratio between first and second be 2:3 and between second and third be 5:8, then the second number is:
- A. 30
B. 20
C. 58
D. 48
10. The average age of A, B, C, D five years ago was 45 years. By including x, the present average age of all the five is 49 years. The present age of x is:
- A. 64 years
B. 48 years
C. 45 years
D. 40 years
11. The average age of 8 men is increased by 2 years. When 2 of them, whose ages are 20 years and 24 years respectively, are replaced by 2 women. What is the average age of these two women?

Analytical Skills-I

- A. 36 years
- B. 30 years
- C. 40 years
- D. 42 years

12. At a language school, each student is given a score that measures his or her fluency. The fluency score is a weighted average that is determined by rating the student on a scale of 0 to 10 in three categories: grammar, vocabulary, and pronunciation. Grammar counts for 40% of the score, vocabulary counts for 25%, and pronunciation counts for 35%. Thomas gets ratings of 8 for grammar, 6 for vocabulary, and 5 for pronunciation. What is his fluency score?

- A. 7.45
- B. 6.45
- C. 7.50
- D. 7.70

13. A grade point average is a weighted average that gives greater weight to courses that earn more credits. Hailey's grade points are 4.0 in Chemistry, which is worth 4 credits, 3.5 in English, which is worth 3 credits, and 3.7 in Physics, which is worth 2 credits. What is Hailey's grade point average?

- A. 8.65
- B. 3.76
- C. 9.88
- D. 7.70

14. The 30 students in Ms. Chen's class had an average grade of 85 on a standardized test. The 20 students in Mr. Jackson's class had an average grade of 90 on the same test. What is the average test grade of all 50 students?

- A. 80.0
- B. 60
- C. 19.5
- D. 67.20

15. A batsman makes a score of 87 runs in the 17th inning and thus increases his avg by. Find his average after 17th inning.

- A. 39
- B. 40
- C. 42
- D. 69

Answers for Self Assessment

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. C | 2. D | 3. A | 4. C | 5. A |
| 6. A | 7. C | 8. B | 9. A | 10. C |
| 11. B | 12. B | 13. B | 14. D | 15. A |

Review Questions

1. There were 35 students in a hostel. Due to the admission of 7 new students, the expenses of the mess were increased by Rs. 42 per day while the average expenditure per head diminished by Rs 1. What was the original expenditure of the mess?
2. The average weight of 10 oarsmen in a boat is increased by 1.8 kg when one of the crew, who weighs 53 kg is replaced by a new man. Find the weight of the new man.
3. The average age of a class of 39 students is 15 years. If the age of the teacher be included, then the average increases by 3 months. Find the age of the teacher.
4. The average weight of A,B,C is 45 Kg. The avg wgt of A & B be 40Kg & that of B,C be 43Kg. Find the wgt of B.
5. Find the average of first 40 natural numbers?
6. A tourist covers half of this journey by train at 60 km/h, half of the remainder by bus at 30 km/h and the rest by cycle at 10 km/h. What is the average speed of the tourist in km/h during his entire journey.
7. What is the average of all multiples of 10 from 2 to 198?
8. The average weight of 16 boys in a class is 50.25 kg and that of the remaining 8 boys is 45.15 kg. Find the average weights of all the boys in the class.
9. In what ratio must water be mixed with milk to gain $16\frac{2}{3}\%$ on selling the mixture at cost price?
10. In what ratio must a grocer mix two varieties of pulses costing H15 and H20 per kg respectively so as to get a mixture worth H16.50 kg?



Further Readings

1. Quantitative Aptitude For Competitive Examinations By Dr. R S Aggarwal, S Chand Publishing.
2. A Modern Approach To Verbal & Non-Verbal Reasoning By Dr. R S Aggarwal, S Chand Publishing
3. Magical Book On Quicker Maths By M Tyra, Banking Service Chronicle
4. Analytical Reasoning By M.K. Pandey, Banking Service Chronicle